

AUTONOMOUS MOBILE ROBOT · MULTI-DOMAIN SYSTEM

Line Following, Wall Detection & Ball Carriage

A self-contained robot integrating mechanical, electronic and control design to complete a timed multi-stage course.

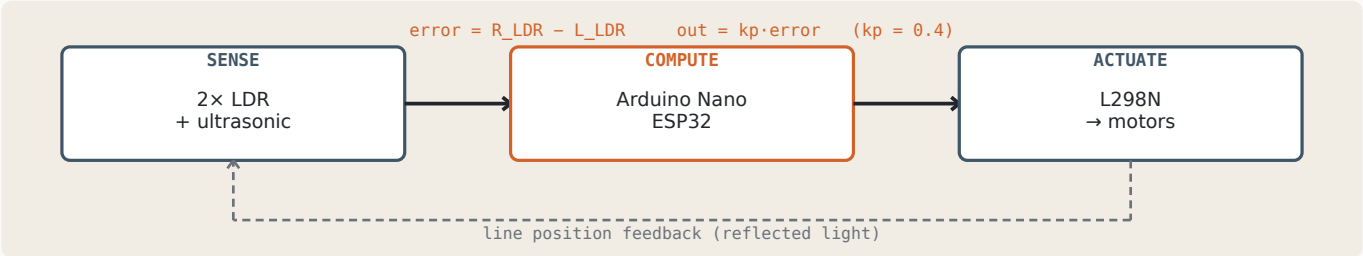


Fig. 1 – Sense → compute → actuate loop. A tuned proportional controller drives differential steering from a two-LDR line sensor; an ultrasonic sensor halts the robot within 30 cm of the wall.

ROLE	COVERS	STACK
Team project (3 members). Contributed across chassis design, sensor electronics and control-software tuning.	Chassis & layout · sensor head · control system · engineering decisions · honest outcome	Arduino Nano ESP32 · L298N driver · LDR dividers · HC-SR04 · SolidWorks + laser-cut acrylic · embedded C

Brief

The task was to design, build and test an autonomous robot that follows a curved black line on a white surface, stops within 30 cm of a wall using a proximity sensor, reverses into a designated parking area, and carries a ball throughout the course without dropping it — all within a 220 mm cube envelope. The robot was realised as a tightly integrated mechanical, electronic and control system, with weight distribution, sensor positioning and control simplicity prioritised for reliable performance.

Result at a glance

kp = 0.4
CONTROL GAIN

15 m
LINE COMPLETED

<30 cm
WALL STOP

0
BALL DROPS

The core tasks — line following, wall stop and ball carriage — were all completed to a functional standard. The reversing/parking stage was the main gap (see Outcome on the following page).

System specifications

SUBSYSTEM	DETAIL
Controller	Arduino Nano ESP32 (240 MHz dual-core)
Motor driver	L298N dual H-bridge, PWM speed control
Drive	2× DC gear motors, differential, rear-mounted
Caster	3D-printed front roller ball
Line sensor	2× LDR + LED, voltage dividers on A0/A1, 35 mm spacing
Proximity	HC-SR04 ultrasonic, time-of-flight
Chassis	200 × 150 mm acrylic, 3 mm laser-cut
Envelope	≤ 220 mm cube (design limit)
Control law	Proportional, $k_p = 0.4$, base PWM 100
Wall-stop	< 30 cm threshold → 0 PWM both motors
Firmware	Embedded C (Arduino framework)

Chassis & layout

A compact 200 × 150 mm acrylic plate was designed in SolidWorks and laser-cut at 3 mm — ample room for all components while staying well inside the envelope. Two drive motors are mounted symmetrically at the rear for differential steering; a 3D-printed roller-ball caster at the front keeps the chassis level, holds consistent ground clearance for the sensors, and lowers the centre of mass for cornering stability.

Electronics (Arduino, L298N, battery) sit centrally, concentrating weight over the driven axle to maximise traction. A 100 mm front slit lets the sensor head be adjusted without a redesign, and the ball cup sits at the centre-rear pivot point to minimise the lateral acceleration the ball experiences while turning.

Sensor head

Line position is sensed by two LDRs with two illuminating LEDs on a 3D-printed plate, the LDRs spaced 35 mm either side of centre. Reflected-light intensity drops over the black line, raising LDR resistance; each LDR forms a voltage divider read on analogue pins A0/A1. Centred, both LDRs read equally; on drift, the imbalance gives a signed error that drives the steering correction. A 40 mm foam shield blocks ambient light for consistent sensing.

Control system

The Arduino reads both LDRs and computes a signed error, $\text{error} = R_LDR - L_LDR$, capturing both the direction and magnitude of deviation. A proportional law converts this to a steering correction, $\text{out} = k_p \cdot \text{error}$, applied differentially: $\text{left} = \text{base} + \text{out}$, $\text{right} = \text{base} - \text{out}$, with base speed 100 leaving PWM headroom for correction. This is smoother than binary on/off logic. The gain $k_p = 0.4$ was found empirically — below 0.3 was sluggish and lost the line on tight

curves; above 0.6 over-corrected and weaved.

An HC-SR04 ultrasonic sensor measures wall distance via pulse time-of-flight; below 30 cm, both motors are commanded to zero PWM, meeting the stopping requirement before the parking stage.

Control & signal parameters

PARAM	VALUE
Error signal	$e = R_LDR - L_LDR$ (signed, 10-bit ADC)
Proportional gain	$k_p = 0.4$ (tuned; usable 0.3–0.6)
Output law	$\text{left} = 100 + k_p \cdot e$; $\text{right} = 100 - k_p \cdot e$
PWM range	0–255 (8-bit), base 100
LDR spacing	±35 mm from centreline
Ambient shield	40 mm foam hood over sensor head
Wall threshold	300 mm (HC-SR04, ~58 $\mu\text{s}/\text{cm}$)

Engineering decisions

Several alternatives were evaluated and deliberately rejected to fit the schedule. A three-LDR sensor was dropped because its more complex algorithm needed calibration time the timetable did not allow; the two-sensor error proved sufficient and far quicker to tune. Underneath- and rear-mounted battery layouts were rejected for poor packaging and a centre-of-mass shift that cut traction. Extra ball-retention features (side supports, a pendulum) were tested but rejected once the egg-cup geometry alone proved sufficient — reducing mechanical complexity.

Outcome

The robot completed the line-following, wall-stop and ball-carriage tasks reliably. The clearest engineering lesson came from the encoder loop: each motor carried a quadrature encoder intended to close an inner speed loop and equalise wheel speeds against motor tolerance and battery sag. When the encoder inner loop and the LDR outer loop ran together, interaction between the two layers introduced oscillation that worsened line following.

Engineering judgement: rather than ship an unstable two-layer controller, the encoder correction was disabled for the run and the robot relied on the thoroughly-tuned, demonstrably-stable proportional controller. The reversing/parking task — which depended on encoder distance counting — was the remaining incomplete objective. The documented next step is a full PID inner loop, where the derivative term damps the oscillation and the integral term removes steady-state offset.

Skills demonstrated

Mechatronic system integration; embedded C on Arduino; analogue sensor interfacing (LDR voltage dividers, ultrasonic time-of-flight); proportional feedback control and empirical gain tuning; mechanical design in SolidWorks with laser-cut and 3D-printed parts; and the engineering judgement to ship a stable, well-characterised subsystem under time constraint rather than an unproven one.

Planned development

The clear next iteration is to re-enable the quadrature encoders inside a properly tuned PID inner loop: the derivative term to damp the cross-coupling oscillation observed with the outer LDR loop, and the integral term to remove the steady-state wheel-speed mismatch from motor tolerance and battery sag. With reliable encoder distance counting restored, the reversing and parking objective becomes achievable through dead-reckoning, and the differential drive gains closed-loop speed regulation independent of supply voltage.